Context:

- Target is a globally renowned brand and a prominent retailer in the United States. Target makes itself a preferred shopping destination by offering outstanding value, inspiration, innovation and an exceptional guest experience that no other retailer can deliver.
- This particular business case focuses on the operations of Target in Brazil and provides insightful information about 100,000 orders placed between 2016 and 2018. The dataset offers a comprehensive view of various dimensions including the order status, price, payment and freight performance, customer location, product attributes, and customer reviews.

Problem Statement:

- Extracting valuable insights and providing actionable recommendations from the given data

Data Description:

The data is available in 8 different csv files:

1. customers.csv
2. geolocation.csv
3. order_items.csv
4. payments.csv
5. reviews.csv
6. orders.csv
7. products.csv
8. Sellers.csv

Analysis

I. **Import the dataset and do usual exploratory analysis steps like checking the structure & characteristics of the dataset:**

A. **Data type of all columns in the "customers" table.**

```
SELECT
       column_name,
       data_type
FROM
      `project.dataset.INFORMATION_SCHEMA.COLUMNS`
WHERE
      table_name = 'customers'
```

column_name	data_type
customer_id	STRING
customer_unique_id	STRING
customer_zip_code_prefix	INT64
customer_city	STRING
customer_state	STRING

* INFORMATION_SCHEMA.COLUMNS is a special function in bigquery.

Insights:

- In the customers table, we have 5 columns, in which 4 are of String data type and 1 is integer.

B. **Get the time range between which the orders were placed.**

A. During what time of the day, do the Brazilian customers mostly place their orders? (Dawn, Morning, Afternoon or Night)
    - 0-6 hrs : Dawn
    - 7-12 hrs : Mornings
    - 13-18 hrs : Afternoon
    - 19-23 hrs : Night

Solution SQL
Screenshot

Insights:

- Max. orders are placed in Afternoon, and lowest orders are placed in morning.

Recommendations:

- In the afternoon, have promos/discounts rolled out and show advertisements.
- Have sufficient stock/inventory during peak hours.

Suggestions (Good Practices):

- **Try to write clean-readable queries**
- **Have proper indentation**
- **Use CTEs for large queries**
- **Provide insights and recommendations wherever possible.**